

Thermodynamic Criticality in FLRW Cosmology with Non-extensive Loop Quantum Gravity Entropy

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We investigate thermodynamic criticality in a spatially flat Friedmann-Lemaître-Robertson-Walker universe with a non-extensive Loop Quantum Gravity inspired entropy on its apparent horizon. Using the full Kodama-Hayward temperature and the unified first law, we derive the modified Friedmann dynamics and construct the corresponding horizon equation of state. We see that a finite physical critical point only appears in the non-extensive branch $q > 1$, while in the Bekenstein-Hawking limit $q \rightarrow 1$, the critical point is continuously pushed to $v_c \rightarrow \infty$ and $T_c \rightarrow 0$. Along the same spinodal curve, both the constant-pressure heat capacity and the isothermal compressibility become unbounded, and the extremum of this curve is located at the critical point. Below the critical temperature, the Gibbs free energy develops several thermodynamic branches, and the phase coexistence for different horizon states is verified by the equality of temperature, pressure, and Gibbs free energy. The critical exponents are $(\alpha_{cr}, \beta_{cr}, \gamma_{cr}, \delta_{cr}) = (0, 1/2, 1, 3)$, indicating that the system belongs to the standard mean-field universality class despite the non-algebraic form of the equation of state. The normalized Ruppeiner curvature diverges precisely on the spinodal curve and shows critical scaling $R_N \sim -|v - v_c|^{-4}$ on the critical isotherm, and $R_N \sim -|t|^{-2}$ on the critical isochore. Finally, the critical expansion scale is given by $H_c^2 = 2(\sqrt{5} - 2)|\beta|$, which implies that the thermodynamic critical point occurs when the entropy deformation is of order unity. These results establish a self-consistent critical structure for the effective thermodynamic state space of the cosmological apparent horizon and retain the mean-field critical universality.

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I. Introduction

Among the most suggestive hints of microscopic degrees of freedom of spacetime are those arising from the relationship between gravitation and thermodynamics. The laws of black hole mechanics, the Bekenstein-Hawking area law and Hawking radiation, provided a consistent assignment of entropy and temperature to gravitational horizons. Subsequently, this relation was generalized to the dynamics of spacetime itself, including the derivation of gravitational field equations from thermodynamic relations [1–6]. In cosmological spacetimes, the apparent horizon provides a natural boundary to define thermodynamic quantities. Hayward’s unified first law is a thermodynamic description of a Friedmann-Lemaître-Robertson-Walker (FLRW) universe. In this formulation, the matter sector is included by means of the work density and the energy-supply vector, and the Kodama-Hayward temperature is determined by the dynamical surface gravity [7–11]. The construction does not require the apparent horizon to be stationary. In particular, keeping the full Kodama-Hayward temperature retains the contribution related to the dynamical evolution of the horizon and is therefore suitable for the study of truly dynamical thermodynamic phenomena.

A major generalization of this framework is the replacement of the Bekenstein-Hawking entropy by a generalized entropy. The motivation for these modifications comes from non-extensive statistical mechanics, quantum-gravity considerations, fractal horizon structures, and corrections to the microscopic counting of horizon states. As representative examples, we mention the Tsallis, Rényi, Barrow, Kaniadakis, and Loop Quantum Gravity motivated entropies [12–20]. Ascribing such entropy functionals to the cosmological apparent horizon modifies the thermodynamic derivation of the Friedmann equations, leading to effective cosmological dynamics that differ from those of standard General Relativity.

Within this more general framework, Majhi showed that non-extensive statistical mechanics can be consistently embedded in the microscopic Loop Quantum Gravity formulation of black-hole entropy [21]. On the basis of this construction, Lymperis wrote down the corresponding LQG-inspired entropy in the exponential non-extensive form that is used in the present work and studied the implications for cosmology [22]. This entropy thus provides a concrete arena in which a quantum-geometry-inspired change of horizon microphysics can be propagated into the macroscopic dynamics of an expanding universe.

A complementary development is that of thermodynamic phase transitions in gravitational systems. Charged AdS black holes have Van der Waals-like isotherms, first-order phase transitions, spinodal instabilities, divergent response functions, and mean-field critical exponents in extended black hole thermodynamics [23, 24]. In more recent times, similar constructions have been developed for cosmological apparent horizons [25–32]. In this context, the thermodynamic pressure is identified with the work density, and the specific volume is taken to be proportional to the apparent-horizon radius. A particularly important result is that the usual Einstein-FLRW thermodynamics does not have a finite P - V critical point [25]. However, nontrivial critical configurations can be produced by modified theories of gravity or generalized horizon entropies [28–30]. This leads us to the main question, which is the subject of the present work: can the non-extensive LQG-inspired entropy give rise to real thermodynamic criticality on the cosmological apparent horizon while the Bekenstein-Hawking system has no finite critical point? This question is nontrivial in the sense that the LQG-inspired entropy is exponentially dependent on the area of the horizon. The corresponding Friedmann dynamics and horizon equation of state are thus non-algebraic, with exponential-integral contributions. This entropy has been previously considered in terms of the cosmological dynamics [22], but not, to our knowledge, systematically investigated in terms of its full thermodynamic phase structure at the apparent horizon.

In this work we tackle this problem by means of the full dynamical Kodama-Hayward formalism. We obtain the modified Friedmann equations from the unified first law without using a quasi-static or adiabatic approximation. Then we identify the thermodynamic pressure with the work density and introduce the specific volume $v = 2R_A$ to formulate the exact horizon equation of state. We analytically demonstrate that a finite positive-temperature critical point only exists in the non-extensive branch $q > 1$. In the Bekenstein-Hawking limit, the critical point is moved to the infinite specific volume and zero critical temperature, which is consistent with the absence of finite criticality in the standard Einstein-FLRW thermodynamics [25]. The heat capacities, the isothermal compressibility, the spinodal structure, and the Gibbs free energy are analyzed, and phase coexistence between different horizon thermodynamic states is established. It is shown that the critical exponents are the usual mean-field values $(\alpha_{cr}, \beta_{cr}, \gamma_{cr}, \delta_{cr}) = (0, 1/2, 1, 3)$. We further investigate the thermodynamic geometry by means of the normalized Ruppeiner curvature. Its singular locus coincides with the spinodal instability, and its critical divergence is directly related to the divergence of the isothermal compressibility. The combined analysis reveals a clear separation between global and local critical properties: the entropy deformation controls the existence and location of the critical point, while the local analytic structure of the equation of state retains the standard mean-field universality class.

The paper is structured as follows. In Sec. II we introduce the non-extensive LQG-inspired entropy on the cosmological apparent horizon. In Sec. III we obtain the modified Friedmann equations from the unified first law. The horizon equation of state and its Bekenstein-Hawking limit are discussed in Sec. IV. We then determine the critical point and study the response functions, spinodal structure, Gibbs free energy and critical exponents. The cosmological

significance of the critical scale is discussed in Sec. X and the thermodynamic geometry is explored in Sec. XI. We conclude by summarizing our results and discussing their physical implications.

II. Non-extensive Loop Quantum Gravity Inspired Entropy

We apply the non-extensive entropy proposed by Majhi and then applied to cosmology by Lymperis, motivated by Loop Quantum Gravity, to include the quantum-geometric corrections in the thermodynamics of the cosmological apparent horizon. The construction unites the microscopic counting of horizon states in Loop Quantum Gravity [33, 34] with a non-extensive statistical description, which in turn entails an exponential deformation of the Bekenstein-Hawking area law. The entropy is given by

$$S_{\text{LQG}} = \frac{1}{1-q} \left[e^{(1-q)\Lambda S_{\text{BH}}} - 1 \right], \quad (1)$$

where $S_{\text{BH}} = A/4G$ is the Bekenstein-Hawking entropy, q is the non-extensive entropic index, and $\Lambda = \Lambda(\gamma_0)$ is a dimensionless quantity determined by the Barbero-Immirzi parameter γ_0 . The value of this parameter is fixed by the microscopic counting of quantum-geometry horizon states, which reproduces the Bekenstein-Hawking area law [33, 34]. In the underlying Loop Quantum Gravity construction,

$$\Lambda(\gamma_0) = \frac{\ln 2}{\sqrt{3}\pi\gamma_0}. \quad (2)$$

For the canonical choice $\gamma_0 = \ln 2/(\pi\sqrt{3})$, one obtains $\Lambda = 1$. In the extensive limit $q \rightarrow 1$, Eq. (1) reduces to

$$\lim_{q \rightarrow 1} S_{\text{LQG}} = \Lambda S_{\text{BH}}. \quad (3)$$

Thus, for the canonical normalization $\Lambda = 1$, the standard Bekenstein-Hawking area law is recovered. For a spatially flat FLRW universe, the apparent-horizon radius is $R_A = 1/H$ and its area is $A = 4\pi R_A^2$. Consequently,

$$S_{\text{BH}} = \frac{\pi R_A^2}{G}, \quad (4)$$

and it is convenient to introduce

$$\beta = \frac{\pi(1-q)\Lambda}{G}. \quad (5)$$

The entropy then takes the compact form $S_{\text{LQG}} = (1-q)^{-1} (e^{\beta R_A^2} - 1)$. For $G > 0$ and $\Lambda > 0$, the sign of β is determined entirely by the non-extensive parameter q , namely $\beta < 0$ if and only if $q > 1$. This condition is not simply a convenient choice of parameters, as we show below. This is a direct consequence of the requirement that the critical specific volume be real and positive. Thus, the non-extensive entropy $q > 1$ branch is selected by thermodynamic criticality. This compact form of the entropy provides the thermodynamic input for the cosmological construction in the following sections. The modification of Friedmann dynamics due to the unified first law is determined by its derivative in relation to the apparent-horizon radius.

III. Modified Friedmann Equations

Using the unified first law at the apparent horizon, we now obtain the cosmological dynamics induced by the non-extensive Loop Quantum Gravity motivated entropy. For a spatially flat FLRW universe, the entropy introduced in Sec. II can be written as

$$S_{\text{LQG}} = \frac{1}{1-q} (e^{\beta R_A^2} - 1), \quad \beta = \frac{\pi(1-q)\Lambda}{G}. \quad (6)$$

Differentiating with respect to the apparent-horizon radius gives

$$dS_{\text{LQG}} = \frac{2\pi\Lambda R_A}{G} e^{\beta R_A^2} dR_A. \quad (7)$$

The gravitational field equations can be written in the unified first-law form [7, 9, 10]. The unified first law projected along the apparent horizon is written in Clausius form as

$$\delta Q = -dE + W dV = T dS_{\text{LQG}}, \quad (8)$$

where $W = \frac{1}{2}(\rho - p)$ is the work density,

$$V = \frac{4\pi}{3} R_A^3 \quad (9)$$

is the volume enclosed by the apparent horizon, and

$$E = \rho V = \frac{4\pi}{3} \rho R_A^3 \quad (10)$$

is the Misner-Sharp energy. The energy enclosed by the apparent horizon is identified with the Misner-Sharp energy [7, 35]. Taking the time derivative of Eq. (10) yields

$$\dot{E} = 4\pi R_A^2 \rho \dot{R}_A + \frac{4\pi}{3} R_A^3 \dot{\rho}, \quad (11)$$

while

$$\dot{V} = 4\pi R_A^2 \dot{R}_A. \quad (12)$$

Therefore,

$$-\dot{E} + W\dot{V} = -2\pi R_A^2 (\rho + p) \dot{R}_A - \frac{4\pi}{3} R_A^3 \dot{\rho}. \quad (13)$$

Using the continuity equation

$$\dot{\rho} + 3H(\rho + p) = 0, \quad (14)$$

the heat flow can be expressed as

$$-\dot{E} + W\dot{V} = 4\pi H R_A^3 (\rho + p) \left(1 - \frac{\dot{R}_A}{2H R_A} \right). \quad (15)$$

For a dynamical apparent horizon, the appropriate temperature is associated with the Kodama-Hayward surface gravity [7, 36],

$$\kappa = -\frac{1}{R_A} \left(1 - \frac{\dot{R}_A}{2H R_A} \right). \quad (16)$$

We adopt the corresponding positive horizon temperature

$$T = -\frac{\kappa}{2\pi} = \frac{1}{2\pi R_A} \left(1 - \frac{\dot{R}_A}{2H R_A} \right). \quad (17)$$

No quasi-static approximation is imposed in Eq. (17). From Eq. (7),

$$\dot{S}_{\text{LQG}} = \frac{2\pi \Lambda R_A}{G} e^{\beta R_A^2} \dot{R}_A, \quad (18)$$

and therefore

$$T \dot{S}_{\text{LQG}} = \frac{\Lambda}{G} e^{\beta R_A^2} \dot{R}_A \left(1 - \frac{\dot{R}_A}{2H R_A} \right). \quad (19)$$

Substitution of Eqs. (15) and (19) into Eq. (8) shows that the full dynamical factor appearing in the Kodama-Hayward temperature occurs identically on both sides of the Clausius relation and therefore cancels exactly. One obtains

$$\frac{\Lambda}{G} e^{\beta R_A^2} \dot{R}_A = 4\pi H R_A^3 (\rho + p). \quad (20)$$

Using Eq. (14), this relation becomes

$$\dot{\rho} = -\frac{3\Lambda}{4\pi G} \frac{e^{\beta R_A^2}}{R_A^3} \dot{R}_A, \quad (21)$$

or equivalently

$$d\rho = -\frac{3\Lambda}{4\pi G} \frac{e^{\beta R_A^2}}{R_A^3} dR_A. \quad (22)$$

Integrating Eq. (22) gives

$$\rho(R_A) = \frac{3\Lambda}{8\pi G} \left[\frac{e^{\beta R_A^2}}{R_A^2} - \beta \text{Ei}(\beta R_A^2) \right] + \rho_0, \quad (23)$$

where ρ_0 is an integration constant. In the following we set $\rho_0 = 0$, since a nonzero value can be absorbed into an independent vacuum-energy contribution. Using $R_A = 1/H$, Eq. (23) becomes

$$\rho(H) = \frac{3\Lambda}{8\pi G} \left[H^2 e^{\beta/H^2} - \beta \text{Ei}\left(\frac{\beta}{H^2}\right) \right]. \quad (24)$$

Equation (24) is the modified first Friedmann equation generated by the non-extensive entropy. Its non-algebraic structure follows directly from the exponential dependence of the entropy on the apparent-horizon area. As a consistency check, consider the extensive limit $q \rightarrow 1$, $\beta \rightarrow 0$. Since $\beta \text{Ei}(\beta/H^2) \rightarrow 0$, Eq. (24) reduces to

$$\rho = \frac{3\Lambda}{8\pi G} H^2. \quad (25)$$

For the canonical normalization $\Lambda = 1$, one therefore recovers the standard Friedmann equation,

$$H^2 = \frac{8\pi G}{3} \rho. \quad (26)$$

IV. Equation of State

We now construct the thermodynamic equation of state associated with the cosmological apparent horizon. Following the thermodynamic formulation of the FLRW apparent horizon, we identify the work density with the thermodynamic pressure,

$$P = W = \frac{\rho - p}{2}, \quad (27)$$

as considered in Refs. [25, 26]. Since

$$P = \rho - \frac{1}{2}(\rho + p), \quad (28)$$

the equation of state can be obtained once the combination $\rho + p$ is expressed in terms of the horizon temperature and radius. From Eq. (20),

$$\frac{\Lambda}{G} e^{\beta R_A^2} \dot{R}_A = 4\pi H R_A^3 (\rho + p). \quad (29)$$

The Kodama-Hayward temperature,

$$T = \frac{1}{2\pi R_A} \left(1 - \frac{\dot{R}_A}{2H R_A} \right), \quad (30)$$

gives

$$\dot{R}_A = 2H R_A (1 - 2\pi R_A T). \quad (31)$$

Consequently,

$$\rho + p = \frac{\Lambda}{2\pi G R_A^2} e^{\beta R_A^2} (1 - 2\pi R_A T). \quad (32)$$

Using Eq. (23) with the integration constant fixed to $\rho_0 = 0$, Eqs. (28) and (32) yield

$$P(T, R_A) = \frac{\Lambda T}{2G R_A} e^{\beta R_A^2} + \frac{\Lambda}{8\pi G R_A^2} e^{\beta R_A^2} - \frac{3\Lambda\beta}{8\pi G} \text{Ei}(\beta R_A^2). \quad (33)$$

To facilitate comparison with fluid and black-hole thermodynamics, we introduce the specific volume $v = 2R_A$ and define

$$x = \frac{\beta v^2}{4}. \quad (34)$$

The exact equation of state then becomes

$$P(T, v) = \frac{\Lambda}{G} e^x \left(\frac{T}{v} + \frac{1}{2\pi v^2} \right) - \frac{3\Lambda\beta}{8\pi G} \text{Ei}(x). \quad (35)$$

The central thermodynamic relation of the model is given by Eq. (35). The temperature dependent part is modulated by the exponential entropy deformation, while the exponential-integral term is induced by the modified Friedmann dynamics. Their combination yields a non-algebraic equation of state, which exhibits a phase structure qualitatively different from the standard FLRW apparent horizon. The Bekenstein-Hawking limit provides an immediate consistency check. For $q \rightarrow 1$, $\Lambda \rightarrow 1$ and $\beta \rightarrow 0$, one has $e^x \rightarrow 1$ and $\beta \text{Ei}(x) \rightarrow 0$, so that Eq. (35) reduces to

$$P(T, v) = \frac{T}{v} + \frac{1}{2\pi v^2}. \quad (36)$$

This is the standard equation of state of the apparent horizon in a spatially flat FLRW universe. As shown in the following section, Eq. (36) does not possess a finite critical point, whereas the non-extensive equation of state Eq. (35) does so for an appropriate branch of the deformation parameter. This result is consistent with the absence of a finite critical point in the standard Einstein-FLRW thermodynamic description [25]. For completeness, retaining the integration constant ρ_0 in Eq. (23) would add a constant term to the pressure,

$$P(T, v) \rightarrow P(T, v) + \rho_0. \quad (37)$$

Such a term does not affect the criticality conditions involving derivatives with respect to v and thus leaves v_c and T_c unchanged. However, it shifts the critical pressure. For the rest of this work, we adopt the vacuum-energy normalization of the thermodynamic pressure by setting $\rho_0 = 0$. The representative isotherms of Eq. (35) around the critical temperature are shown in Fig. 1. For $T < T_c$, a local maximum and a local minimum appear, i.e., there are multiple thermodynamic branches separated by a mechanically unstable region. At $T = T_c$ the two extrema merge into one inflection point and for $T > T_c$ the isotherms are monotonic. The qualitative structure of the equation of state is illustrated in Fig. 1, where the emergence of a characteristic Van der Waals-like oscillatory behavior below the critical temperature can already be identified.

V. Critical Point

The critical point is determined by the standard inflection-point conditions on the isotherm [23, 37].

$$\left(\frac{\partial P}{\partial v} \right)_T = 0, \quad \left(\frac{\partial^2 P}{\partial v^2} \right)_T = 0. \quad (38)$$

For the equation of state (35), the first derivative is

$$\left(\frac{\partial P}{\partial v} \right)_T = \frac{\Lambda e^x}{2\pi G v^3} (\pi T \beta v^3 - 2\pi T v - \beta v^2 - 2), \quad (39)$$

where $x = \beta v^2/4$, as defined in Eq. (34). The second derivative is

$$\left(\frac{\partial^2 P}{\partial v^2} \right)_T = \frac{\Lambda e^x}{4\pi G v^4} (\pi T \beta^2 v^5 - 2\pi T \beta v^3 + 8\pi T v - \beta^2 v^4 + 12). \quad (40)$$

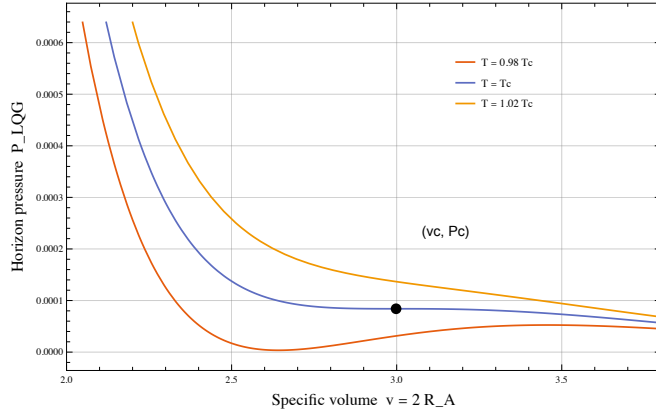


Figure 1: Isotherms of the LQG-FLRW equation of state for temperatures $T/T_c = 0.98, 1.00$, and 1.02 . At $T = T_c$, the point (v_c, P_c) is an inflection point of the critical isotherm. For $T < T_c$, a local maximum and minimum appear, defining multiple thermodynamic branches separated by a mechanically unstable region. For $T > T_c$, the equation of state becomes monotonic.

Eliminating the temperature between Eqs. (39) and (40), and defining

$$y = \beta v_c^2, \quad (41)$$

one obtains

$$y^2 + 8y - 4 = 0. \quad (42)$$

The two algebraic roots are

$$y_{\pm} = -4 \pm 2\sqrt{5}. \quad (43)$$

The temperature obtained from Eq. (39) is

$$T_c = \frac{y + 2}{\pi v_c (y - 2)}. \quad (44)$$

For the root

$$y_+ = -4 + 2\sqrt{5} > 0, \quad (45)$$

the requirement $v_c^2 > 0$ implies $\beta > 0$, but Eq. (44) then gives $T_c < 0$. Therefore this branch does not correspond to a physical positive-temperature critical point. The physical branch is instead $y_- = -4 - 2\sqrt{5} < 0$, for which $v_c^2 > 0$ requires $\beta < 0$. Using $\beta = \pi(1 - q)\Lambda/G$, with $G > 0$ and $\Lambda > 0$, this condition is equivalent to $q > 1$. Hence, a positive-temperature critical point exists only in the non-extensive branch $q > 1$. The critical specific volume is

$$v_c^2 = \frac{-4 - 2\sqrt{5}}{\beta}, \quad (46)$$

while the critical temperature can be written in the compact form

$$T_c = \frac{\sqrt{5} - 1}{2\pi v_c}. \quad (47)$$

At the critical point,

$$x_c = \frac{\beta v_c^2}{4} = -1 - \frac{\sqrt{5}}{2}, \quad (48)$$

and the critical pressure follows directly from Eq. (35),

$$P_c = \frac{\Lambda}{G v_c^2} \left[\frac{\sqrt{5}}{2\pi} e^{-1-\sqrt{5}/2} + \frac{3(2+\sqrt{5})}{4\pi} \text{Ei} \left(-1 - \frac{\sqrt{5}}{2} \right) \right]. \quad (49)$$

For the representative choice $q = 1.3$, $\Lambda = 1$ and $G = 1$, one obtains

$$v_c \simeq 2.99820, \quad T_c \simeq 0.065615, \quad P_c \simeq 8.4013 \times 10^{-5}. \quad (50)$$

For the canonical choice $\Lambda = G = 1$, the dimensionless critical ratio takes the numerical value

$$Z_c = \frac{P_c v_c}{T_c} \simeq 3.8389 \times 10^{-3}. \quad (51)$$

The critical isotherm develops an inflection point at (v_c, P_c) . For $T < T_c$, the equation of state possesses two stationary points and therefore multiple thermodynamic branches, whereas for $T > T_c$ the isotherms are monotonic.

VI. Thermodynamic Response Functions

Thermodynamic response functions provide information about the local behavior of the horizon system and become singular at the boundaries of thermodynamic stability. We consider the heat capacities at constant volume and constant pressure, together with the isothermal compressibility. The heat capacity at constant volume is defined as

$$C_V = T \left(\frac{\partial S}{\partial T} \right)_V. \quad (52)$$

Since

$$V = \frac{\pi v^3}{6}, \quad (53)$$

fixing the thermodynamic volume fixes v and therefore the apparent-horizon radius. Because the entropy depends only on the horizon radius,

$$S_{\text{LQG}} = \frac{1}{1-q} \left(e^{\beta v^2/4} - 1 \right), \quad (54)$$

it follows immediately that $C_V = 0$. This result will imply the critical exponent $\alpha_{cr} = 0$. The heat capacity at constant pressure is

$$C_P = T \left(\frac{\partial S}{\partial T} \right)_P. \quad (55)$$

At fixed pressure,

$$dP = \left(\frac{\partial P}{\partial T} \right)_v dT + \left(\frac{\partial P}{\partial v} \right)_T dv = 0, \quad (56)$$

and hence

$$\left(\frac{dT}{dv} \right)_P = - \frac{\left(\frac{\partial P}{\partial v} \right)_T}{\left(\frac{\partial P}{\partial T} \right)_v}. \quad (57)$$

Using

$$\frac{dS_{\text{LQG}}}{dv} = \frac{\pi \Lambda v}{2G} e^x, \quad \left(\frac{\partial P}{\partial T} \right)_v = \frac{\Lambda}{Gv} e^x, \quad (58)$$

together with Eq. (39), one obtains

$$C_P = - \frac{\pi^2 \Lambda T v^3 e^x}{G (\pi T \beta v^3 - 2\pi T v - \beta v^2 - 2)}. \quad (59)$$

It is useful to define

$$D(T, v) = \pi T \beta v^3 - 2\pi T v - \beta v^2 - 2. \quad (60)$$

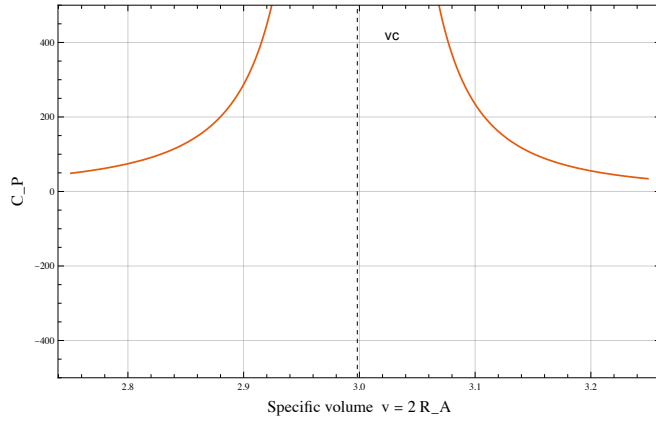


Figure 2: Heat capacity at constant pressure C_P evaluated along the critical isotherm $T = T_c$. The singularity at $v = v_c$ follows from the condition $(\partial P/\partial v)_T = 0$, which coincides with the spinodal locus.

Since

$$\left(\frac{\partial P}{\partial v}\right)_T = \frac{\Lambda e^x}{2\pi G v^3} D(T, v), \quad (61)$$

the heat capacity diverges whenever $D(T, v) = 0$. Now, the isothermal compressibility is defined by

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P}\right)_T. \quad (62)$$

Using $V = \pi v^3/6$, one obtains

$$\kappa_T = -\frac{3}{v} \left[\left(\frac{\partial P}{\partial v}\right)_T \right]^{-1}, \quad (63)$$

and therefore

$$\kappa_T = -\frac{6\pi G v^2 e^{-x}}{\Lambda (\pi T \beta v^3 - 2\pi T v - \beta v^2 - 2)}. \quad (64)$$

Equations (59) and (64) show that both C_P and κ_T possess the same singular locus,

$$D(T, v) = 0 \quad \Longleftrightarrow \quad \left(\frac{\partial P}{\partial v}\right)_T = 0. \quad (65)$$

This locus is precisely the spinodal curve discussed in the next section. The critical point belongs to this curve and is distinguished by the additional condition

$$\left(\frac{\partial^2 P}{\partial v^2}\right)_T = 0. \quad (66)$$

The sign of the isothermal compressibility provides the direct criterion for local mechanical stability. Regions with $\kappa_T > 0$ are mechanically stable, whereas $\kappa_T < 0$ corresponds to a mechanically unstable branch. The divergences of C_P and κ_T therefore identify the boundary separating the different local thermodynamic branches. The divergence of the response functions along the spinodal curve is the standard signature of the loss of local thermodynamic stability [37]. Figures 2 and 3 illustrate the behavior of the response functions along the critical isotherm. Both quantities become singular at $v = v_c$, consistently with the analytic conditions above.

VII. Spinodal Curve

The spinodal curve determines the boundary of local mechanical stability of the horizon system. Along this curve the isothermal compressibility diverges, or equivalently,

$$\left(\frac{\partial P}{\partial v}\right)_T = 0. \quad (67)$$

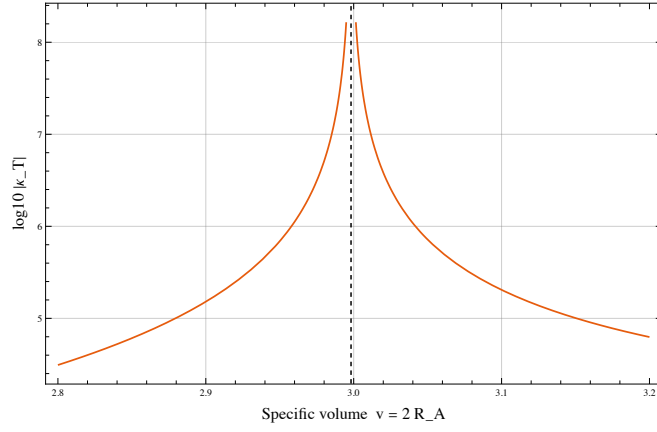


Figure 3: Logarithmic representation of the isothermal compressibility, $\log_{10} |\kappa_T|$, along the critical isotherm $T = T_c$. The divergence at $v = v_c$ corresponds to the critical point of the spinodal curve.

Using Eq. (39), the spinodal condition becomes

$$\pi T \beta v^3 - 2\pi T v - \beta v^2 - 2 = 0. \quad (68)$$

Solving for the temperature yields the exact spinodal curve

$$T_{\text{sp}}(v) = \frac{\beta v^2 + 2}{\pi v(\beta v^2 - 2)}. \quad (69)$$

The critical point corresponds to the extremum of the spinodal temperature. Differentiating Eq. (69) gives

$$\frac{dT_{\text{sp}}}{dv} = -\frac{\beta^2 v^4 + 8\beta v^2 - 4}{\pi v^2(\beta v^2 - 2)^2}. \quad (70)$$

Therefore,

$$\frac{dT_{\text{sp}}}{dv} = 0 \quad (71)$$

implies

$$\beta^2 v^4 + 8\beta v^2 - 4 = 0. \quad (72)$$

This is exactly the same equation obtained from the simultaneous criticality conditions

$$\left(\frac{\partial P}{\partial v}\right)_T = \left(\frac{\partial^2 P}{\partial v^2}\right)_T = 0. \quad (73)$$

Hence the extremum of the spinodal curve coincides analytically with the thermodynamic critical point. For the physical branch,

$$\beta v_c^2 = -4 - 2\sqrt{5}, \quad (74)$$

and substitution into Eq. (69) gives

$$T_{\text{sp}}(v_c) = T_c = \frac{\sqrt{5} - 1}{2\pi v_c}. \quad (75)$$

The corresponding pressure along the spinodal is obtained parametrically from

$$P_{\text{sp}}(v) = P(T_{\text{sp}}(v), v). \quad (76)$$

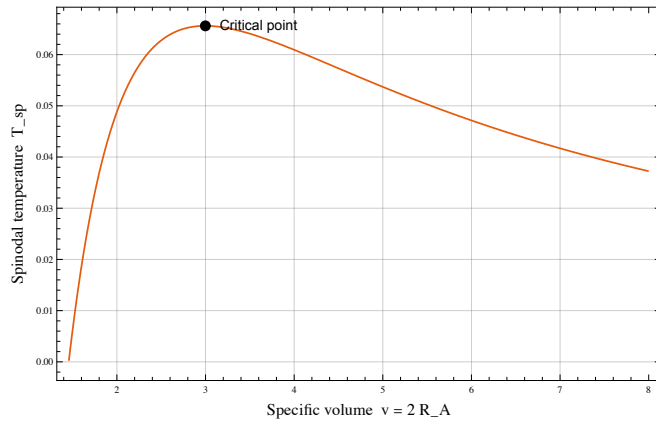


Figure 4: Spinodal curve in the (T, v) plane for the non-extensive LQG-FLRW cosmological model. The curve is defined by $(\partial P / \partial v)_T = 0$, and its maximum coincides analytically with the critical point (v_c, T_c) . For $T < T_c$, the two branches of the spinodal delimit the mechanically unstable region in which $\kappa_T < 0$.

Equations (69) and (76) therefore provide a parametric representation of the spinodal boundary in the (P, T) plane. For temperatures below the critical value, the equation

$$T = T_{\text{sp}}(v) \quad (77)$$

admits two solutions,

$$v_-(T) < v_+(T), \quad (78)$$

which delimit the mechanically unstable branch. In this intermediate region $\kappa_T < 0$, whereas the outer branches satisfy $\kappa_T > 0$. The spinodal curve therefore characterizes local mechanical stability. The distinction between globally stable and metastable branches requires the Gibbs free-energy analysis presented in the next section. Figure 4 shows the spinodal curve in the (T, v) plane. Its maximum occurs at (v_c, T_c) , where the two spinodal branches merge. For $T < T_c$, the two intersections of a horizontal isotherm with the spinodal define the limits of the mechanically unstable region.

Since

$$\kappa_T \propto D^{-1}, \quad (79)$$

whereas

$$R_N \propto -D^{-2}, \quad (80)$$

the leading critical behavior satisfies

$$|R_N| \propto \kappa_T^2. \quad (81)$$

Using $\gamma_{cr} = 1$, this is consistent with

$$R_N \propto -|t|^{-2\gamma_{cr}} = -|t|^{-2}. \quad (82)$$

VIII. Gibbs Free Energy

The local stability analysis based on the response functions and the spinodal curve does not determine which of the locally stable thermodynamic branches is globally preferred. This information is encoded in the Gibbs free energy. The thermodynamic construction employed below refers to the effective state space of the apparent horizon, with the work density playing the role of thermodynamic pressure, $P = W$. Accordingly, phase coexistence should be understood as coexistence between distinct horizon thermodynamic states, rather than as a microscopic material phase transition of the cosmological fluid.

We begin with the unified first law of thermodynamics applied at the apparent horizon,

$$T dS = -dE + W dV, \quad (83)$$

where $E = \rho V$ is the total energy inside the horizon, $W = (\rho - p)/2$ is the work density and $V = 4\pi R_A^3/3$ is the volume enclosed by the apparent horizon. Introducing the internal energy $U = -E$ and identifying the thermodynamic pressure with the work density, Eq. (83) takes the standard thermodynamic form $dU = T dS - P dV$. The identification $U = -E$ therefore follows directly from the horizon first law and does not constitute an additional thermodynamic assumption.

The Gibbs free energy is defined in the usual way and, using $U = -\rho V$, can be written as

$$\mathcal{G} = U + PV - TS = (P - \rho)V - TS. \quad (84)$$

Using $P - \rho = -\frac{1}{2}(\rho + p)$ together with Eq. (32), we obtain

$$(P - \rho)V = \frac{\Lambda e^{\beta R_A^2}}{3G} (2\pi T R_A^2 - R_A). \quad (85)$$

In terms of the specific volume $v = 2R_A$ and the variable $x = \beta v^2/4$ introduced in Sec. IV, the Gibbs free energy becomes

$$\mathcal{G}(T, v) = \frac{\Lambda e^x}{6G} (\pi T v^2 - v) - \frac{T}{1 - q} (e^x - 1). \quad (86)$$

As a consistency check, Eqs. (83) and (84) imply

$$d\mathcal{G} = -S dT + V dP, \quad (87)$$

and therefore

$$\left(\frac{\partial \mathcal{G}}{\partial P} \right)_T = V = \frac{\pi v^3}{6} > 0. \quad (88)$$

This relation provides a direct thermodynamic consistency check for the parametric Gibbs curves. At fixed temperature, the pressure and Gibbs free energy can be regarded as parametric functions of the specific volume,

$$P = P(T, v), \quad \mathcal{G} = \mathcal{G}(T, v). \quad (89)$$

For $T < T_c$, the nonmonotonic equation of state produces multiple branches in the $\mathcal{G}$ - P plane. The spinodal analysis identifies which of these branches are locally stable or unstable, while global coexistence requires two locally stable states, v_s and v_l , to satisfy simultaneously

$$P(T, v_s) = P(T, v_l) = P_{\text{coex}}, \quad (90)$$

and

$$\mathcal{G}(T, v_s) = \mathcal{G}(T, v_l). \quad (91)$$

The coexistence pressure may equivalently be determined through Maxwell's equal-area construction [23, 38].

$$\int_{V_s}^{V_l} [P(V, T) - P_{\text{coex}}] dV = 0. \quad (92)$$

Since

$$V = \frac{\pi v^3}{6}, \quad (93)$$

this condition becomes

$$\int_{v_s}^{v_l} [P(v, T) - P_{\text{coex}}] \frac{\pi v^2}{2} dv = 0. \quad (94)$$

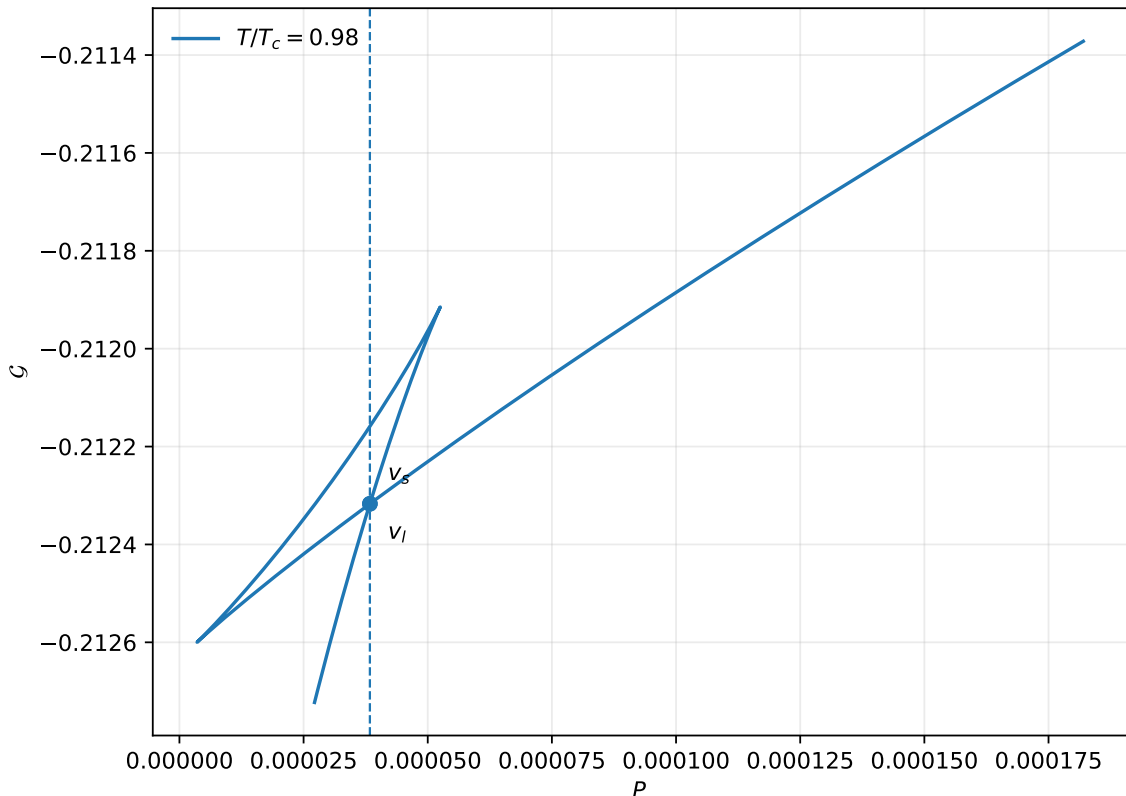


Figure 5: Parametric Gibbs free energy $\mathcal{G}(P)$ at $T = 0.98T_c$ for $q = 1.3$ and $\Lambda = G = 1$. The two marked states, $v_s \simeq 2.43101$ and $v_l \simeq 3.97563$, have the same coexistence pressure, $P_{\text{coex}} \simeq 3.83492 \times 10^{-5}$, and the same Gibbs free energy, $\mathcal{G}_{\text{coex}} \simeq -0.212317$. They therefore represent distinct coexisting thermodynamic states of the apparent horizon.

For the representative choice $q = 1.3$ and $\Lambda = G = 1$, and at $T = 0.98T_c$, the coexistence conditions yield

$$v_s \simeq 2.43101, \quad v_l \simeq 3.97563, \quad (95)$$

with

$$P_{\text{coex}} \simeq 3.83492 \times 10^{-5}, \quad (96)$$

and

$$\mathcal{G}_{\text{coex}} \simeq -0.212317. \quad (97)$$

The two states therefore have the same temperature, pressure, and Gibbs free energy, providing a direct thermodynamic verification of phase coexistence between distinct horizon states. Figure 5 shows this coexistence explicitly. The two marked points correspond to the states v_s and v_l , which share the same values of P_{coex} and $\mathcal{G}_{\text{coex}}$. As $T \rightarrow T_c$, the two coexisting branches approach each other and merge at the critical point.

IX. Critical Exponents and Universality Class

We now determine the critical exponents governing the singular behavior of the thermodynamic variables near the critical point. The critical exponents are defined through the standard asymptotic scaling relations near the critical point [37, 39]

$$t = \frac{T - T_c}{T_c}, \quad \phi = \frac{v - v_c}{v_c}, \quad p = \frac{P - P_c}{P_c}. \quad (98)$$

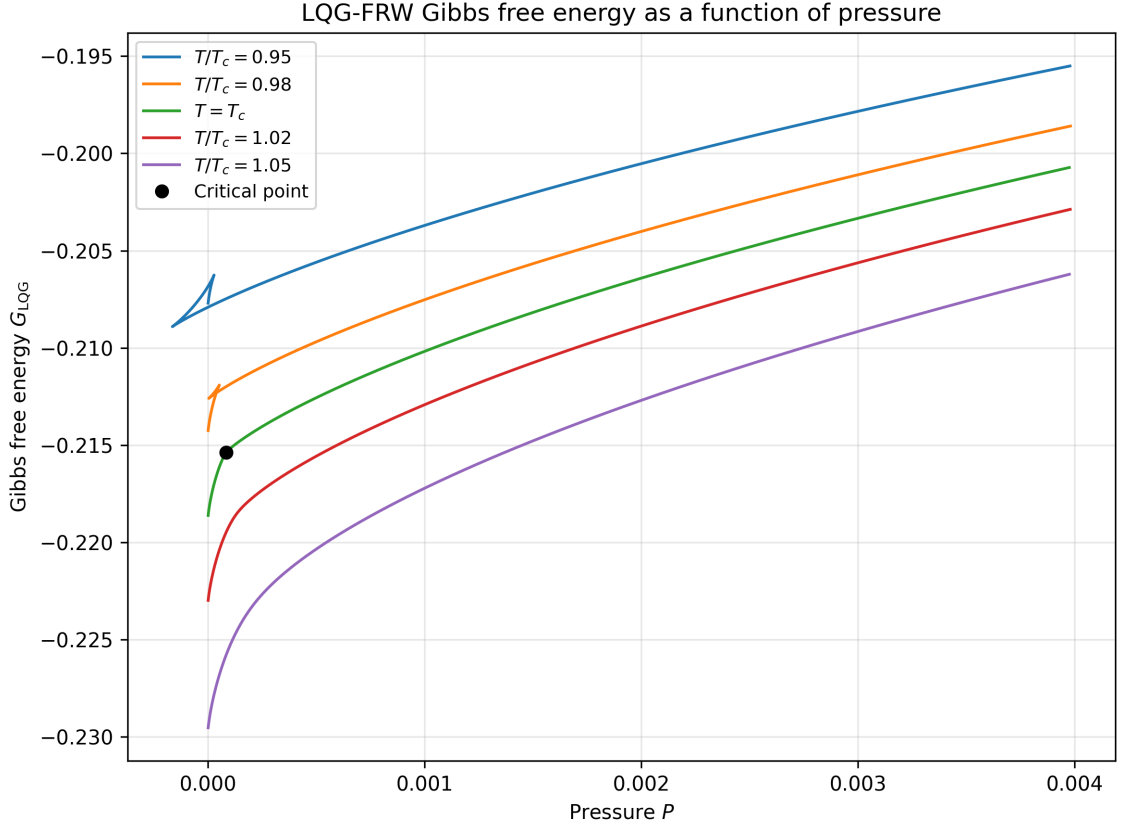


Figure 6: Parametric Gibbs free energy $\mathcal{G}(P)$ for several temperatures around the critical point. For $T < T_c$, the nonmonotonic equation of state produces a multibranch structure associated with locally stable and unstable thermodynamic states. The branches merge as $T \rightarrow T_c$, while for $T > T_c$ the curve becomes single-valued.

Thus,

$$T = T_c(1 + t), \quad v = v_c(1 + \phi). \quad (99)$$

Expanding the equation of state around (T_c, v_c) gives

$$p = a_{10}t + a_{11}t\phi + a_{03}\phi^3 + O(t\phi^2, \phi^4, t^2), \quad (100)$$

where

$$a_{10} = \frac{T_c}{P_c} \left(\frac{\partial P}{\partial T} \right)_c, \quad (101)$$

$$a_{11} = \frac{T_c v_c}{P_c} \left(\frac{\partial^2 P}{\partial T \partial v} \right)_c, \quad (102)$$

and

$$a_{03} = \frac{v_c^3}{6P_c} \left(\frac{\partial^3 P}{\partial v^3} \right)_c. \quad (103)$$

The absence of terms proportional to ϕ and ϕ^2 follows directly from the criticality conditions

$$\left(\frac{\partial P}{\partial v} \right)_c = 0, \quad \left(\frac{\partial^2 P}{\partial v^2} \right)_c = 0. \quad (104)$$

For the physical critical branch, the coefficients a_{11} and a_{03} are both finite and nonzero. For the representative choice $q = 1.3$ and $\Lambda = G = 1$, one finds

$$a_{10} \simeq 31.3287, \quad a_{11} \simeq -164.039, \quad a_{03} \simeq -61.1337. \quad (105)$$

Thus, the critical point is an ordinary cubic critical point. The critical exponent α_{cr} characterizes the behavior of the heat capacity at constant volume,

$$C_V \propto |t|^{-\alpha_{cr}}. \quad (106)$$

Since the entropy depends only on the horizon radius, we found in Sec. VI that $C_V = 0$. Therefore, $\alpha_{cr} = 0$. The order parameter may be defined as the difference between the large- and small-volume branches,

$$\eta = v_l - v_s. \quad (107)$$

Although the physical thermodynamic volume is

$$V = \frac{\pi v^3}{6}, \quad (108)$$

near the critical point one has

$$V - V_c = \frac{\pi v_c^2}{2}(v - v_c) + O((v - v_c)^2), \quad (109)$$

so that using $v_l - v_s$ or $V_l - V_s$ as the order parameter yields the same critical exponent. For $t < 0$, the coexistence conditions applied to Eq. (100) give, at leading order,

$$\phi_l = -\phi_s \propto (-t)^{1/2}. \quad (110)$$

Consequently,

$$\eta \propto (-t)^{1/2}, \quad (111)$$

and therefore

$$\beta_{cr} = \frac{1}{2}. \quad (112)$$

The exponent γ_{cr} describes the divergence of the isothermal compressibility,

$$\kappa_T \propto |t|^{-\gamma_{cr}}. \quad (113)$$

Along the critical isochore $\phi = 0$, Eq. (100) gives

$$\left(\frac{\partial p}{\partial \phi}\right)_t = a_{11}t + O(t^2). \quad (114)$$

Since $a_{11} \neq 0$, it follows that

$$\kappa_T \propto \frac{1}{|t|}, \quad (115)$$

and hence $\gamma_{cr} = 1$. Along the critical isotherm $t = 0$, the leading-order equation of state reduces to

$$p = a_{03}\phi^3 + O(\phi^4). \quad (116)$$

Since $a_{03} \neq 0$,

$$|P - P_c| \propto |v - v_c|^3, \quad (117)$$

and therefore $\delta_{cr} = 3$. The complete set of critical exponents is therefore

$$(\alpha_{cr}, \beta_{cr}, \gamma_{cr}, \delta_{cr}) = \left(0, \frac{1}{2}, 1, 3\right). \quad (118)$$

These are the standard mean-field critical exponents [37, 39]. They satisfy the standard scaling relations

$$\alpha_{cr} + 2\beta_{cr} + \gamma_{cr} = 2, \quad (119)$$

$$\alpha_{cr} + \beta_{cr}(1 + \delta_{cr}) = 2, \quad (120)$$

and

$$\gamma_{cr} = \beta_{cr}(\delta_{cr} - 1). \quad (121)$$

Thus, despite the non-algebraic exponential-integral structure of the equation of state, the critical point belongs to the standard mean-field universality class. The non-extensive entropy deformation determines the existence and thermodynamic location of the critical point, while the local critical scaling remains of ordinary mean-field type. This mean-field behavior is noteworthy because the present equation of state is not of the algebraic Van der Waals form. Instead, it contains both exponential and exponential-integral contributions inherited from the non-extensive LQG-inspired entropy.

The origin of the mean-field behavior can be understood directly from the local structure of the equation of state. At an ordinary critical point,

$$\left(\frac{\partial P}{\partial v}\right)_c = \left(\frac{\partial^2 P}{\partial v^2}\right)_c = 0, \quad (122)$$

while

$$\left(\frac{\partial^2 P}{\partial T \partial v}\right)_c \neq 0, \quad \left(\frac{\partial^3 P}{\partial v^3}\right)_c \neq 0. \quad (123)$$

Consequently, the leading nontrivial expansion around the critical point has the generic form

$$p = a_{10}t + a_{11}t\phi + a_{03}\phi^3 + \dots, \quad (124)$$

which is sufficient to generate the standard mean-field critical exponents. This remark separates two conceptually different aspects of the thermodynamics. The entropy deformation determines the existence of a finite critical point and fixes its position in the thermodynamic parameter space. The local universality class is, however, determined by the analytic structure of the equation of state near that critical point. Hence the present non-extensive LQG-inspired model is an example of a system where ordinary mean-field critical scaling is preserved despite a strongly non-algebraic equation of state. In this sense the microscopic entropy deformation changes the global thermodynamic structure without changing the local critical exponents. Similar mean-field behavior has been reported in several gravitational thermodynamic systems with generalized entropy or modified horizon dynamics [23, 28–30]. However, whether this behavior persists must be established model by model, since the existence of a critical point and the required analyticity conditions are not guaranteed for an arbitrary entropy functional.

X. Cosmological Interpretation of the Critical Scale

The thermodynamic criticality found in the previous sections is induced by the same entropy deformation which modifies the cosmological Friedmann equation. Thus, it is useful to understand how the thermodynamic critical scale relates to the rate of cosmological expansion. The modified Friedmann equation obtained from the non-extensive LQG-inspired entropy is

$$\rho(H) = \frac{3\Lambda}{8\pi G} \left[H^2 e^{\beta/H^2} - \beta \operatorname{Ei} \left(\frac{\beta}{H^2} \right) \right], \quad (125)$$

where

$$\beta = \frac{\pi(1-q)\Lambda}{G}. \quad (126)$$

The physical critical branch requires $q > 1$, and therefore $\beta < 0$. The departure from the standard Friedmann dynamics is controlled by the dimensionless combination

$$x_H = \frac{\beta}{H^2}. \quad (127)$$

Consequently, the characteristic scale associated with the entropy deformation is

$$H_\beta \sim \sqrt{|\beta|}. \quad (128)$$

For

$$\frac{|\beta|}{H^2} \ll 1, \quad (129)$$

the entropy deformation is small and the dynamics approaches the Bekenstein-Hawking regime. By contrast, when

$$H^2 \sim |\beta|, \quad (130)$$

the deformation becomes of order unity, and a perturbative expansion around the Bekenstein-Hawking limit is no longer adequate. The thermodynamic specific volume is related to the Hubble parameter through

$$v = 2R_A = \frac{2}{H}. \quad (131)$$

Therefore, the critical specific volume determines a corresponding critical expansion scale,

$$H_c = \frac{2}{v_c}. \quad (132)$$

An equivalent characterization follows directly from the dimensionless deformation parameter $x_H = \beta/H^2$. At the critical point,

$$x_{H,c} = \frac{\beta}{H_c^2} = -\frac{1}{2(\sqrt{5}-2)} = -\frac{2+\sqrt{5}}{2} \simeq -2.118. \quad (133)$$

Thus, the critical point lies outside the small-deformation regime $|x_H| \ll 1$. In particular, the phase transition cannot be captured by a perturbative expansion around the Bekenstein-Hawking limit. The critical behavior emerges only when the entropy deformation becomes dynamically significant. Using

$$v_c^2 = \frac{-4 - 2\sqrt{5}}{\beta}, \quad (134)$$

and the physical branch $\beta < 0$, we obtain

$$H_c^2 = 2(\sqrt{5}-2)|\beta|, \quad (135)$$

or equivalently

$$H_c = \sqrt{2(\sqrt{5}-2)}\sqrt{|\beta|} \simeq 0.6871\sqrt{|\beta|}. \quad (136)$$

The critical point therefore does not introduce an independent cosmological scale. Instead, it occurs at an expansion rate set directly by the entropy-deformation scale. This relation becomes particularly transparent by evaluating the dimensionless deformation variable at the critical point. Since $x_H = \beta/H^2 = \beta v^2/4$, one finds $x_c = \beta/H_c^2 = -1 - \sqrt{5}/2 \simeq -2.11803$, and hence $|\beta/H_c^2| \simeq 2.11803$. The critical point therefore lies outside the perturbative regime $|\beta|/H^2 \ll 1$. The phase transition is intrinsically associated with the nonperturbative sector of the entropy deformation. The extensive limit provides another useful interpretation. As $q \rightarrow 1$, one has $\beta \rightarrow 0$, and consequently

$$v_c \rightarrow \infty, \quad T_c \rightarrow 0, \quad H_c \rightarrow 0. \quad (137)$$

Hence the finite thermodynamic critical point is continuously removed in the Bekenstein-Hawking limit by being pushed to the degenerate infrared boundary of the thermodynamic state space. This is in accordance with the absence of finite criticality in the standard FLRW equation of state. However, it is essential to distinguish the existence of a thermodynamic critical point from the realization of one on a physical cosmological trajectory. A cosmological solution must fulfill the modified Friedmann equation, the continuity equation and the equation of state of the matter sector. Thus, existence of (T_c, v_c, P_c) in the thermodynamic state space does not ensure the crossing of this point in the cosmological evolution. To determine if the critical regime can be reached in a realistic cosmological history one needs to specify the matter content and solve for $H(z)$. The present analysis determines the thermodynamic structure and identifies the corresponding expansion scale H_c ; whether this critical regime is dynamically accessible is a separate cosmological question. Remarkably, $x_{H,c}$ is independent of the particular values of q , Λ , and G ; these parameters determine the physical critical scale through β , while the reduced deformation at criticality is fixed entirely by the criticality conditions.

XI. Thermodynamic Geometry and Ruppeiner Curvature

Ruppeiner geometry provides a geometric formulation of equilibrium thermodynamics in which the scalar curvature of the thermodynamic state space can be used as a diagnostic of thermodynamic interactions and critical behavior [40–43]. Thermodynamic geometry provides an independent characterization of the stability and critical structure obtained from the equation of state. In the Ruppeiner formulation, the scalar curvature of the thermodynamic state space is commonly interpreted as an effective measure of thermodynamic interactions and correlations. For the present horizon system $C_V = 0$, and the standard two-dimensional Ruppeiner metric becomes degenerate. For systems with $C_V = 0$, it is useful to introduce the normalized thermodynamic curvature, which remains finite away from the instability locus and retains the relevant information about the critical behavior [44–46]. We therefore consider the normalized Ruppeiner curvature

$$R_N = C_V R_R, \quad (138)$$

which remains finite in the formal $C_V \rightarrow 0$ construction and retains the singular structure of the thermodynamic geometry. For an equation of state linear in the temperature, the normalized curvature can be written as

$$R_N = \frac{1}{2} - \frac{T^2}{2} \left[\frac{\partial^2 P / \partial T \partial v}{(\partial P / \partial v)_T} \right]^2. \quad (139)$$

For the equation of state

$$P(T, v) = \frac{\Lambda}{G} e^x \left(\frac{T}{v} + \frac{1}{2\pi v^2} \right) - \frac{3\Lambda\beta}{8\pi G} \text{Ei}(x), \quad x = \frac{\beta v^2}{4}, \quad (140)$$

one obtains

$$\left(\frac{\partial P}{\partial v} \right)_T = \frac{\Lambda e^x}{2\pi G v^3} D(T, v), \quad (141)$$

where

$$D(T, v) = \pi T \beta v^3 - 2\pi T v - \beta v^2 - 2, \quad (142)$$

and

$$\frac{\partial^2 P}{\partial T \partial v} = \frac{\Lambda e^x}{2G v^2} (\beta v^2 - 2). \quad (143)$$

Substitution into Eq. (139) yields

$$R_N(T, v) = \frac{1}{2} - \frac{\pi^2 T^2 v^2 (\beta v^2 - 2)^2}{2D(T, v)^2}. \quad (144)$$

The overall factor Λ/G cancels identically. The normalized curvature therefore depends on the thermodynamic variables and on the entropy deformation through β , but not explicitly on the overall normalization of the equation of state. Equation (141) immediately implies

$$D(T, v) = 0 \quad \Longleftrightarrow \quad \left(\frac{\partial P}{\partial v} \right)_T = 0. \quad (145)$$

Hence, provided that the numerator of Eq. (144) does not vanish simultaneously,

$$D(T, v) \rightarrow 0 \quad \implies \quad R_N \rightarrow -\infty. \quad (146)$$

The singular locus of the thermodynamic curvature therefore coincides exactly with the spinodal curve. This provides an independent geometric characterization of the mechanical instability boundary. The critical point is distinguished from a generic point of the spinodal by the additional condition

$$\left(\frac{\partial^2 P}{\partial v^2} \right)_c = 0. \quad (147)$$

Since

$$\left(\frac{\partial P}{\partial v}\right)_T = A(v)D(T, v), \quad A(v) = \frac{\Lambda e^x}{2\pi G v^3}, \quad (148)$$

the criticality conditions imply

$$D(T_c, v_c) = 0, \quad \left.\frac{\partial D}{\partial v}\right|_c = 0. \quad (149)$$

Consequently, along the critical isotherm,

$$D(T_c, v) = \frac{1}{2} \left.\frac{\partial^2 D}{\partial v^2}\right|_c (v - v_c)^2 + \dots. \quad (150)$$

Using

$$\left.\frac{\partial^2 D}{\partial v^2}\right|_c = -\frac{2(5 + \sqrt{5})}{v_c^2}, \quad (151)$$

one obtains

$$D(T_c, v) \simeq -\frac{5 + \sqrt{5}}{v_c^2} (v - v_c)^2. \quad (152)$$

Moreover, the numerator of Eq. (144) remains finite and nonzero at the critical point. In fact,

$$\frac{\pi^2 T_c^2 v_c^2}{2} (\beta v_c^2 - 2)^2 = 4(3 + \sqrt{5}). \quad (153)$$

Therefore, the leading critical behavior is

$$R_N(T_c, v) \simeq -\frac{2}{5} \left(\frac{v - v_c}{v - v_c}\right)^4. \quad (154)$$

Equivalently, in terms of the reduced volume variable

$$\phi = \frac{v - v_c}{v_c}, \quad (155)$$

the critical divergence takes the compact form

$$R_N(T_c, v) \simeq -\frac{2}{5} \phi^{-4}. \quad (156)$$

A complementary scaling law follows along the critical isochore. Since $D(T, v)$ is linear in T ,

$$D(T, v_c) = \left.\frac{\partial D}{\partial T}\right|_c (T - T_c), \quad (157)$$

with

$$\left.\frac{\partial D}{\partial T}\right|_c = \pi v_c (\beta v_c^2 - 2) = -2\pi v_c (3 + \sqrt{5}), \quad (158)$$

one obtains

$$R_N(T, v_c) \simeq -\frac{1}{\pi^2 v_c^2 (3 + \sqrt{5})} \frac{1}{(T - T_c)^2}. \quad (159)$$

Introducing the reduced temperature

$$t = \frac{T - T_c}{T_c}, \quad (160)$$

and using

$$T_c = \frac{\sqrt{5} - 1}{2\pi v_c}, \quad (161)$$

the previous expression simplifies to

$$R_N(T, v_c) \simeq -\frac{1}{2t^2}. \quad (162)$$

This result is directly related to the divergence of the isothermal compressibility. Since

$$\kappa_T \propto D^{-1}, \quad (163)$$

whereas

$$R_N \propto -D^{-2}, \quad (164)$$

the leading critical behavior satisfies

$$|R_N| \propto \kappa_T^2. \quad (165)$$

Using $\gamma_{cr} = 1$, this gives

$$R_N \propto -|t|^{-2\gamma_{cr}} = -|t|^{-2}. \quad (166)$$

The normalized curvature is mostly negative away from the singular locus in the thermodynamically stable region. Within the standard Ruppeiner interpretation, negative thermodynamic curvature is commonly associated with effectively attractive interactions [41, 42]. Such an interpretation should be considered as a macroscopic thermodynamic diagnostic, rather than as a microscopic identification of the underlying quantum-gravitational degrees of freedom. Thus, the thermodynamic geometry gives the same stability boundary as obtained independently from the equation of state, response functions and spinodal analysis. More importantly, the critical divergence of R_N obeys definite scaling laws and is directly related to the divergence of the mechanical susceptibility. A complementary view is provided by Fig. 8, where $\log_{10} |R_N|$ is evaluated along the critical isotherm $T = T_c$. The sharp divergence at $v = v_c$ provides a direct numerical verification of the analytical singularity predicted by Eq. (144).

The thermodynamic geometry therefore gives an alternative description of the phase structure obtained from the equation of state, response functions, spinodal curve and Gibbs free energy. In particular the divergence of R_N reproduces the mechanical instability boundary and its singular behavior at the critical point is consistent with the development of long-range correlations. Together with the mean-field critical exponents obtained in the last section, these results indicate that the LQG-corrected apparent horizon exhibits the usual thermodynamic signatures of an interacting critical system.

Thermodynamic geometry has also been applied to the phase structure of FLRW apparent horizons, providing a complementary characterization of cosmological criticality [29, 47]. Future work could involve confronting the modified Friedmann equations with observational data, extending the analysis to non-flat or anisotropic cosmologies, other quantum-gravity-inspired entropy formalisms, and possible connections between thermodynamic phases and the ultimate fate of the universe, such as the de Sitter, Big Rip and Little Rip scenarios.

XII. Conclusions

In this work we have investigated the thermodynamic critical behavior of a spatially flat FLRW universe whose apparent horizon is endowed with a non-extensive LQG-inspired entropy. Starting from the horizon first law and retaining the full Kodama-Hayward temperature, we derived the modified Friedmann dynamics and constructed the corresponding horizon equation of state. The entropy deformation generates a finite critical point only in the non-extensive branch $q > 1$. In the Bekenstein-Hawking limit $q \rightarrow 1$, the critical volume is pushed to infinity while the critical temperature and the corresponding expansion scale vanish. This shows that the finite critical structure is induced by the entropy deformation rather than inherited from standard FLRW horizon thermodynamics. The local and global thermodynamic analyses provide a mutually consistent picture. The heat capacity at constant pressure and the isothermal compressibility diverge along the spinodal curve, whose extremum coincides with the critical point. Below the critical temperature, the Gibbs free energy develops multiple branches, and the coexistence of distinct horizon thermodynamic states is verified by requiring equality of temperature, pressure, and Gibbs free

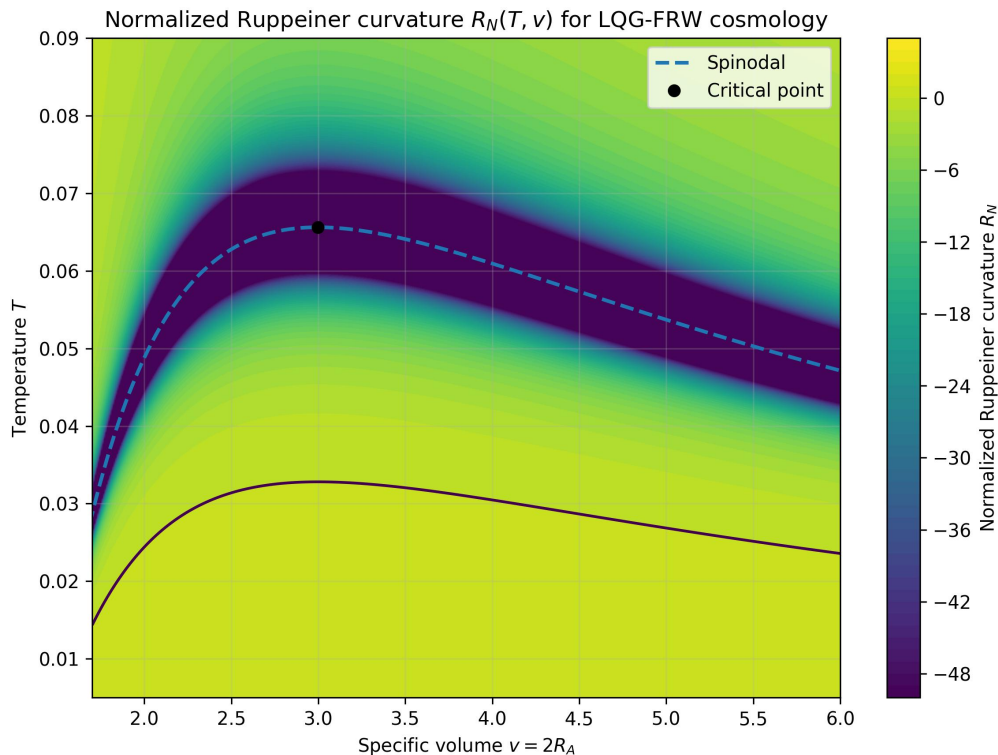


Figure 7: Contour plot of the normalized Ruppeiner curvature $R_N(T, v)$ for the LQG-FLRW cosmological model. The dashed curve denotes the spinodal line defined by $(\partial P / \partial v)_T = 0$, while the black dot marks the critical point (v_c, T_c) . The singular structure of R_N follows the spinodal boundary and the curvature is predominantly negative in the thermodynamically stable region.

energy. This coexistence refers to the effective thermodynamic state space of the apparent horizon and should not be interpreted as a microscopic material phase transition of the cosmological fluid. The critical exponents take the standard mean-field values. Thus, although the entropy deformation determines the existence and location of the critical point, the local critical behavior remains in the mean-field universality class. This separation between the global thermodynamic effects of the entropy deformation and the local universality of the critical point is one of the main results of the present analysis. The normalized Ruppeiner curvature provides an independent geometric characterization of the same structure. Its singular locus coincides with the spinodal curve, and its critical divergence is directly related to the divergence of the isothermal compressibility. The predominantly negative curvature in the stable region is consistent, within the usual Ruppeiner interpretation, with effectively attractive thermodynamic interactions, without implying a specific microscopic identification of the underlying quantum-gravitational degrees of freedom. The critical point also defines a characteristic cosmological expansion scale controlled by the same entropy-deformation parameter that modifies the Friedmann equation. The critical regime lies outside the small-deformation limit, indicating that the phase transition cannot be captured as a perturbative correction around the Bekenstein-Hawking regime. It is important, however, to distinguish thermodynamic criticality in the horizon state space from its realization along a physical cosmological trajectory. The existence of a critical point does not imply that an evolving universe necessarily reaches or crosses it. Addressing this question requires specifying the matter sector and solving the modified cosmological dynamics. Determining whether the critical regime is dynamically accessible and whether it can leave observable signatures in the expansion or perturbation histories constitutes a natural direction for future work.

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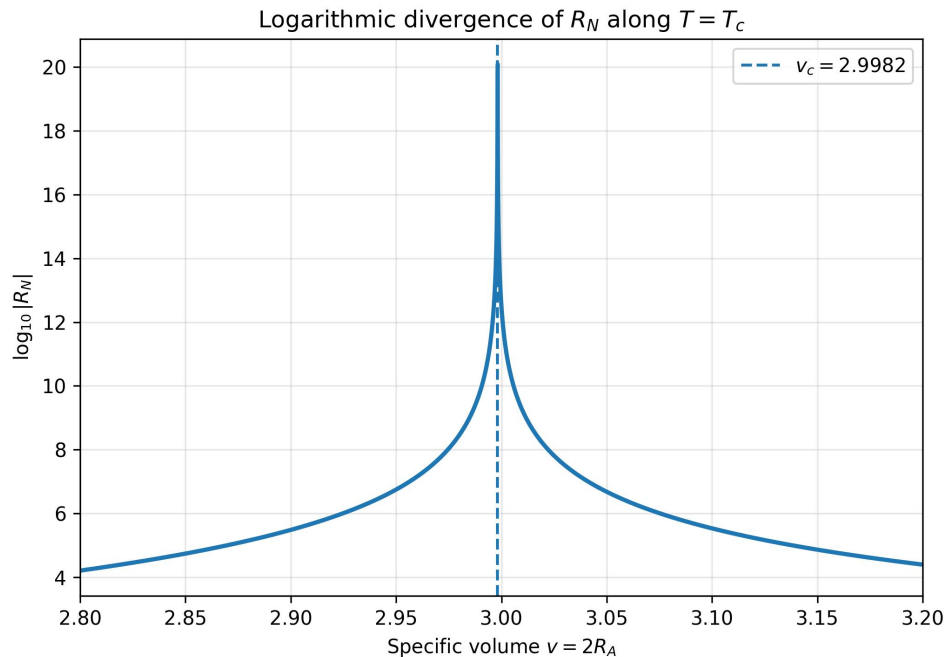


Figure 8: Behavior of $\log_{10} |R_N|$ along the critical isotherm $T = T_c$. The divergence at $v = v_c$ corresponds to the critical point where the two branches of the spinodal curve merge.

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